# Jaya Sharma

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## Introduction

I'm a PhD student in Cybersecurity Engineering at the Technical University of Denmark (DTU), working under the supervision of Dr. Christian Majenz. My PhD research lies at the intersection of theoretical cryptography and quantum computing, with a focus on the security of cryptographic constructions in the quantum random oracle model. So far, I have worked on the security of non-interactive zero-knowledge proofs and on post-quantum threshold signatures based on the hardness of group-based assumptions. My other specific interests include lattice-based cryptography, quantum information and quantum error correction.

## Education

**Technical University of Denmark (DTU)** 2024–27
PhD Student, Cybersecurity Engineering Section, DTU Compute
***Supervisor:*** Dr. Christian Majenz

**Indraprastha Institute of Information Technology Delhi (IIITD)** 2022–24
PhD Junior Research Fellow in the Department of Computer Science and Engineering
***Area:*** Lattice-Based Cryptography and Quantum Computing

**Indian Institute of Technology Roorkee (IITR)** 2020–22
Master of Science in Mathematics
***Grade:*** 8.675/10; All India Rank 122 of 30,942 (top 0.4%) in Joint CSIR-UGC national PhD fellowship exam.

**University of Delhi – Lady Shri Ram College for Women (LSR)** 2017–20
Bachelor of Science (Honours) in Mathematics
***Grade:*** 9.081/10; All India Rank 143 of 14,374 (top 1%) in IIT JAM (national MSc entrance exam).

## Research Visits/Internships

**Centrum Wiskunde & Informatica (CWI), Amsterdam** 09/2026–01/2027
***Host:*** Dr. Serge Fehr
– Working on security proofs for post-quantum non-interactive zero-knowledge protocols.

**QWorld Association** – QIntern *(remote internship)* 07–09/2025
– Worked in an international team of four on a library (currently under review by the Qrisp developers) for efficient block-encoding of dense matrices via QRAM, using the Qrisp framework.

**Indian Institute of Science, Bangalore (IISc)** 06–07/2023
***Host:*** Dr. Chaya Ganesh
– Studied the compression theory of sigma protocols and lattice-based sigma protocols.

**Indian Institute of Science Education and Research Bhopal** *(remote internship)* 06–08/2021
***Host:*** Dr. Sanjay Singh
– Studied an introduction to algebraic function fields and applications in coding theory

## Publications/Theses

**Security of the Fischlin Transform in Quantum Random Oracle Model** 2026
*QCrypt'26 (contributed talk), IACR ASIACRYPT'26- Outstanding Paper*
Christian Majenz, Jaya Sharma

**Elliptic Curves over Complex Numbers** 2022
*Master's Thesis, IITR*; Supervisor: Dr. A. Swaminathan

**Brent–Dekker Method: An Attempt to Decode** 2020
*Eclat – A Mathematics Journal, Vol. 11*
Dr. Monika Singh, Shagun Agrawal, Jaya Sharma

## Presentations

**Fiat–Shamir Transform in QROM** – Crypto Reading Group, DTU 10/2024
**Lattice-Based Sigma Protocols** – Theory Reading Group, IIITD 10/2023

**Trapdoors for Hard Lattices and New Cryptographic Constructions** – Theory Reading Group, IIITD 05/2023
**A Survey on Quantum Error-Correcting Codes** – Introduction to Quantum Computing, IIITD 05/2023

## Academic Activities

**Short Term Visit- Dr. Russell Lai , Aalto University, Finland** 11/2026
**Sub-Reviewer, Eurocrypt'27** 2026
**PhD Committee Member, DTU Compute** 2025
**Teaching Assistant**
– Applied Cryptography, Discrete Mathematics, Graph Theory (DTU) 2024–25
– Introduction to Cryptography (IIT Delhi) 2023
– Discrete Mathematics (IIITD) 2022

## Honours, Awards & Grants

**IACR Asiacrypt Outstanding Paper Award** 2026

**Thomas B. Thriges Fond Grant, William Demant Fonden Grant, Otto Mønsteds Fond Grant** 2026
– Secured three Danish grants to fund a research stay at CWI Amsterdam.

**ACM-India Anveshan Setu Fellowship** 2023
– Awarded by the Association for Computing Machinery (ACM) India to work at IISc.

**Joint CSIR-UGC Junior Research Fellowship** 2022
– Awarded a PhD research fellowship in Mathematical Sciences through the national exam conducted jointly by the Council of Scientific and Industrial Research (CSIR) and the University Grants Commission (UGC), India.

## Workshops/Summer Schools/Conferences

**Asiacrypt– Hong Kong** 12/2026
**ACM CCS– Young Scholar's Development Program- The Hague, Netherlands** 11/2026
**QCrypt – Ottawa, Canada** 08/2026
**Eurocrypt – Rome, Italy** 05/2026
**DQC Scientific Quantum Conference 2026 – Aalborg University, Denmark** 04/2026
**Theory of Cryptography Conference (TCC) 2025 – Aarhus, Denmark** 12/2025
**Workshop on Mathematics of Post-Quantum Cryptography – University of Zurich** 06/2025
**Quantum Safe Internet (QSI) Workshop – Technical University of Denmark, Lyngby** 05/2025
**IACR Summer School on Post-Quantum Cryptography – University of Warsaw** 07/2024
**ACM India Grad Cohort – IISc Bangalore** 07/2023
**CSE Research Symposium – IIT Bombay** 03/2023
**Vigyan Vidushi Mathematics Summer School – TIFR Mumbai** 07/2021
**Vigyan Vidushi Computer Science Summer School – TIFR Mumbai** 06/2021

## Other

**Programming:** C++, Python, Qiskit
**Languages:** English, Hindi (mother tongue), Danish (elementary)